

Moduli Space Reading Group

Week 7: Review

Defⁿ 2.8: Naive Moduli Problem:

Collection $\mathcal{A}$ of objects and an equivalence relation $\sim$ on $\mathcal{A}$.

Examples:

(i) $\mathcal{A} = k$ -dim subspaces of an n -dim vect space
 $\sim = "="$

(ii) $\mathcal{A} =$ Collection of vector bundles on a fixed scheme X .
 $\sim =$ isomorphism of vector bundles.

Motivation: We want to classify objects not just up to equivalence/isom, but we want them to behave in families.

Definition 2.10: Extended Moduli Problem.

Given a naive moduli problem $(\mathcal{A}, \sim)$, an extended moduli problem is given by:

(1) $\forall$ scheme S , a collection $\mathcal{A}_S$ of "families over S ", and an eq relation $\sim_S$ on $\mathcal{A}_S$

(2) for $f: T \rightarrow S$ in Sch , a pull-back $f^*: \mathcal{A}_S \rightarrow \mathcal{A}_T$.

This data must satisfy:

(i) $(\mathcal{A}_{\text{Spec } k}, \sim_{\text{Spec } k}) = (\mathcal{A}_S, \sim_S)$.

(ii) $\text{id}^* = \text{id}$

(iii) $(g \circ f)^* = f^* \circ g^*$

(iv) $f \sim_T g \Rightarrow f^* T \sim_S f^* g$

- For every $s \in S$, the fibre over S gives a family

$$s: \operatorname{Spec} k \longrightarrow S, \quad \mathcal{F} \in \mathcal{A}_s$$

Then $\mathcal{F}_s := s^* \mathcal{F} \rightsquigarrow$ family over $\operatorname{Spec} k$.

Example: Let $V = \mathbb{C}^{n+1}$. Consider

$$\mathcal{A} = \{ L \subseteq V \mid \dim(L) = 1 \}, \quad L \sim L' \Leftrightarrow L = L'.$$

This is a naive moduli problem.

Geometrically: set of lines through origin in $\mathbb{A}^{n+1}$.

As a set: $\mathcal{A} \cong \mathbb{P}^n$.

But we don't know how lines can vary algebraically in a family.

Drawback: Suppose S scheme and $\forall s \in S$, we want a line $L_s \subset V$. We don't want an arbitrary assignment

$$s \longmapsto L_s,$$

we want L_s to vary in some compatibility with variance of s .

This is where extension comes in.

What is the extended moduli problem here?

- For any $S \in \operatorname{Sch}$, define

$$\mathcal{A}_S = \left\{ \begin{array}{l} \text{line bundles over } S \text{ that} \\ \text{are subbundles of trivial} \\ \text{rank } n+1 \text{ vect bundle } V \times S \end{array} \right\}$$

As inv sheaves, element of $\mathcal{A}_S$ looks like $\mathcal{L} \hookrightarrow \mathcal{O}_S^{\oplus n+1}$.

$\sim_S$ is defined to be equality.

• Why does this describe a "family of lines"?

Let $\mathcal{L} \in \mathcal{A}_S$ be a family over S .

- $\forall s \in S$, we have $s: \text{Spec } k \rightarrow S$.

- So every $\mathcal{L} \in \mathcal{A}_S$ pulls back to $s^* \mathcal{L} \subseteq s^*(V \times S) \cong V$.

- And since $\mathcal{L}$ is a line bundle, so $\dim_k \underbrace{s^* \mathcal{L}}_{\text{fiber of the sheaf at } s} = 1$.

- Thus $\mathcal{L}_s := s^* \mathcal{L} \subseteq V$.

- So a single family $\mathcal{L} \subset V \times S$ gives us, for every $s \in S$, an object of the naive moduli problem

$$s \mapsto \mathcal{L}_s \subseteq V.$$

- They vary algebraically:

• For instance, take $V = \mathbb{C}^2$, $S = \mathbb{A}^1 = \text{Spec } \mathbb{C}[t]$.

• Consider the family $\mathcal{L} \subseteq \mathbb{C}^2 \times \mathbb{A}^1$ which at $t=a$ has fiber $\mathcal{L}_a = \text{span} \{ (1, a) \}$.

• So as "a" varies,

$$a=0: \quad \mathcal{L}_0 = \langle (1, 0) \rangle$$

$$a=1: \quad \mathcal{L}_1 = \langle (1, 1) \rangle$$

$$a=2: \quad \mathcal{L}_2 = \langle (1, 2) \rangle$$

$\vdots$

The lines vary algebraically in the expression $(1, t)$ as t varies in the base scheme $S = \text{Spec } \mathbb{C}[t]$.

"Parametrized by S ."

Lemma 2.11: An extended moduli problem $\mathcal{M}$ defines a contravariant functor from schemes to sets as:

$$\mathcal{M}(S) := \mathcal{A}_S / \sim_S = \{ \text{families over } S \} / \sim_S \quad \mathcal{M}(f: T \rightarrow S) := f^*: \mathcal{M}(S) \rightarrow \mathcal{M}(T).$$

Example:

$$\begin{aligned} \mathcal{M}(S) &= \left\{ \begin{array}{l} \text{subbundles } \mathcal{L} \text{ over } S \\ \text{of the trivial bundle} \\ V \times S \end{array} \right\} \quad \left| \quad \begin{array}{l} f: T \rightarrow S \\ \mathcal{M}(f): \mathcal{A}_S \rightarrow \mathcal{A}_T \\ \mathcal{L} \mapsto f^* \mathcal{L} \\ \subseteq V \times S \quad \subseteq f^*(V \times S) \\ \quad \quad \cong V \times T. \end{array} \right. \\ &= \mathcal{A}_S / \sim_S \end{aligned}$$

Remark/Notation: From now on, by a moduli problem we shall mean its moduli functor.

Question: When does a scheme M represent a moduli functor $\mathcal{M}$?

Def^m: If it does, then M is called a fine moduli space for $\mathcal{M}$.

i.e., There is a natural isom

$$\eta: \mathcal{M} \rightarrow \mathcal{h}_M = \text{Hom}(-, M)$$

of functors. This means

- (i) $\forall$ Scheme S , there is a bijection $\eta_S: \mathcal{M}(S) = \mathcal{A}_S / \sim_S \xleftrightarrow{\sim} \mathcal{h}_M(S) := \{ S \rightarrow M \}$
- (ii) $\forall$ morph $T \rightarrow S$, we have:

$$\begin{array}{ccc} \mathcal{A}_S / \sim_S & \xrightarrow{\eta_S} & \{ S \rightarrow M \} \\ \mathcal{M}(f) \downarrow & & \downarrow \\ \mathcal{A}_T / \sim_T & \xrightarrow{\eta_T} & \{ T \rightarrow M \} \end{array} \quad \text{commutes.}$$

- The family in $\mathcal{A}_M / \sim_M = \mathcal{M}(M)$ corresponding to $M \xrightarrow{id} M$ is called the universal family.
- family $\left. \begin{array}{l} \text{For } S \rightarrow f: S \rightarrow M \\ f^* \mathcal{U} \end{array} \right\} f^* \mathcal{U} \sim F$
- Thus, any family is equiv to pull-back of the universal family.
- Fine moduli space is unique upto unique isom.

Examples:

(i) Ex 2.19: $\mathbb{P}^n$ as a fine moduli space.

- Consider the naive moduli problem of lines through the origin in $V = A^{n+1}$

- Extend it in the following way:

A family of lines through the origin in A^{n+1} over S is a line bundle $\mathcal{L}$ over S which is a subbundle of the trivial bundle $V \times S$ over S . Two families are equivalent if the two line bundles are equal.

$$f: S \rightarrow \mathbb{P}^n \longmapsto \mathcal{L} := f^* \mathcal{O}_{\mathbb{P}^n}(-1) \hookrightarrow \mathcal{O}_S^{n+1} = V \times S$$

$$S \longrightarrow \mathbb{P}^n \longleftarrow \mathcal{L}$$

$$s \longmapsto [q \circ \phi(s), \dots, q \circ \phi_n(s)]$$

generating global sections of $\mathcal{L}^\vee$.

$$\therefore \left\{ \text{Morphisms } S \rightarrow \mathbb{P}^n \right\} \longleftrightarrow \left\{ \text{Families of lines through origin in } V \text{ over } S \right\}.$$

Then, $\mathbb{P}^n$ is a fine moduli space and the tautological bundle $\mathcal{O}_{\mathbb{P}^n}(-1)$ is the universal family.

- **Pathological behaviour:** Not every moduli functor has a fine moduli space.

Example: The naive moduli problem of classifying endomorphisms of a n -dimensional vector space.

Extension - A family over S is a rank n vect bundle $\mathcal{F}$ over S with endomorphism $\phi: \mathcal{F} \rightarrow \mathcal{F}$.

This does not admit a fine moduli space.

- **Defⁿ 2.24:** A **coarse moduli space** for a moduli functor $\mathcal{M}$ is a scheme M and a natural transformation $\eta: \mathcal{M} \rightarrow h_M$ such that
 - (a) $\eta_{\text{Spec } k}: \mathcal{M}(\text{Spec } k) \rightarrow h_M(\text{Spec } k)$ is bijective.
 - (b) For any scheme N and a natural transf $v: \mathcal{M} \rightarrow h_N$, there exists a unique morphism of schemes $f: M \rightarrow N$ such that

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{v} & h_N \\ & \searrow & \nearrow \eta_f \\ & h_M & \end{array}$$

Q. When is a coarse moduli space a fine moduli space?

- **Proposition 2.26:** Let (M, η) be a coarse moduli space for a moduli problem $\mathcal{M}$. Then (M, η) is a fine moduli space if and only if
 - (1) there exists a family $\mathcal{U}$ over M such that $\eta_M(\mathcal{U}) = \text{id}_M$.
 - (2) for $\mathcal{F}, \mathcal{G} \in \mathcal{A}_S$: $\mathcal{F} \sim_S \mathcal{G} \iff \eta_S(\mathcal{F}) = \eta_S(\mathcal{G})$.

Next Goal: Construction of moduli spaces:

Objects, equivalence relⁿ

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Parameter space (scheme) P and a family $\mathcal{F}$ over P s.t. locally any family is a pull-back of $\mathcal{F}$

}

Group action on P s.t. $\mathcal{O}(p) = \mathcal{O}(q) \Leftrightarrow \mathcal{F}_p \sim \mathcal{F}_q$.

}

X/G our moduli space.

So this motivates the study of (i) Group actions
(ii) Quotients.

• Defⁿ: Affine alg group G over k :

- Scheme G with morphisms

- $e: \text{Spec } k \rightarrow G$; $m: G \times G \rightarrow G$; $i: G \rightarrow G$ that satisfy the group laws.

• Action of an affine alg group G on a scheme X :

This is a morphism $\sigma: G \times X \rightarrow X$ that is a generalization of group acting on a set.

• Example: $G_m \curvearrowright \mathbb{A}^2$:

Affine alg grp: $G_m := \text{Spec } k[t, t^{-1}]$; Scheme $= \mathbb{A}^2$

Action: $t \cdot (x, y) = (tx, t^{-1}y)$.

Orbits: For any $c \neq 0$, we have the orbit $V_c = V(xy - c)$.

Orbit of $(1,0)$: Punctured x -axis.
Orbit of $(0,1)$: " y -axis.
Orbit of $(0,0)$: $\{(0,0)\}$.

} non-closed orbits b/c $(0,0)$ is in closure

Idea: Insisting that every orbit remain a distinct point can produce a badly behaved quotient. GIT allows certain non-closed orbits to be identified.

• The three notions of Quotients:

G -invariant morphism:

Let $G \curvearrowright X$, $G \curvearrowright Y$ and $f: X \rightarrow Y$, then f is called G -equivariant if f is compatible with the G -action.

Further, if the action on Y is trivial, then f is called G -invariant.

Categorical Quotient for $G \curvearrowright X$:

A G -inv morph $\varphi: X \rightarrow Y$ s.t. every G -inv $f: X \rightarrow Z$ factors uniquely as:

$$\begin{array}{ccc} X & \xrightarrow{f} & Z \\ \searrow \varphi & & \uparrow h \\ & Y & \end{array}$$

So Categorical means universal among all morphisms.

Good Quotient and Geometric Quotient for $G \curvearrowright X$:

Good quotient is a morphism $\varphi: X \rightarrow Y$ if

(i) G -inv (ii) surjective

(iii) The induced action of G on $\mathcal{O}(X)$:

$$g \cdot f(x) = f(g^{-1} \cdot x)$$

gives that $\forall U \subseteq Y$, the morphism $\mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(\psi^{-1}(U))$ is an isom onto the G -inv functions $\mathcal{O}_X(\psi^{-1}(U))^G$.

(iv) $W \subseteq X$ is G -inv + closed $\Rightarrow \psi(W) \subseteq_{\text{closed}} Y$.

(v) $W_1 \cap W_2 = \emptyset$, $W_1, W_2 \subseteq_{\text{closed}} X \Rightarrow \psi(W_1) \cap \psi(W_2) = \emptyset$

(vi) ψ is affine.

Further, if pre-image of each point is a single orbit, then ψ is called a geometric quotient.

Hierarchy:

Geometric Quotient $\Rightarrow$ Good Quotient $\Rightarrow$ Categorical Quotient.

Example: $G_m \curvearrowright A^2$ has the map $\psi: A^2 \rightarrow A^1$ as good quotient
 $(x, y) \mapsto xy$

and hence cat quot. However $\psi^{-1}(0,0) = \text{Union of three orbits}$, so not a geometric quot.

So we see: invariant functions cannot necessarily distinguish all orbits.

This motivates the study of GIT:

Affine GIT:

- Consider an action $G \curvearrowright X = \underbrace{\text{Spec } A}_{\text{finite type over } k}$
- This induces action $G \curvearrowright \mathcal{O}(X) = \text{regular functions}$
 $(g \cdot f)(x) = f(g^{-1} \cdot x).$
- $\mathcal{O}(X)^G := \{ f \in \mathcal{O}(X) \mid g \cdot f = f \}.$

Question: Is $\mathcal{O}(X)^G$ finitely generated k -alg?

Answer: FALSE in general.

TRUE with reductivity hypothesis.

Defⁿ: An affine alg group G is

- (i) linearly reductive if every finite dim lin rep $\rho: G \rightarrow GL(V)$ decomposes completely into direct sum of irred.
- (ii) geometrically reductive if for every finite dim lin rep $\rho: G \rightarrow GL(V)$ and every non-zero G inv point $v \in V$, $\exists$ G -inv non-constant homog poly $f \in \mathcal{O}(V)$ such that $f(v) \neq 0$.

Observation: So geom reductivity is weaker than linear reductivity but strong enough to produce invariant polynomials that separate the configurations needed in invariant theory.

Theorem 4.16: For smooth affine group schemes:

Linearly reductive $\Rightarrow$ geom reductive $\Leftrightarrow$ reductive.

For char $k = 0$, they coincide.

Example: $G = GL_n, SL_n$.

char $k > 0$; GL_n, SL_n are not lin reductive.

(smooth + every smooth unip normal sgp is trivial)